

Variational Autoencoders — Solutions

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

Instructor: Miodrag Bolić, University of Ottawa

February 25, 2026

Part A: Conceptual Questions

A1. Autoencoders vs. Variational Autoencoders

Two fundamental differences:

(1) Encoder output. A deterministic AE encoder is a function $f_\phi : x \mapsto z$; it maps each input to a single point in latent space with no uncertainty. A VAE encoder is a *probabilistic* inference network $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi(x)^2))$; it maps each input to a *distribution* over latent codes, capturing uncertainty about the representation.

(2) Generative capability. A standard AE has no explicit prior $p(z)$. The latent codes of training points may cluster arbitrarily and leave large “holes” in latent space containing no training examples. Sampling a random z from a region not occupied by training data and decoding it produces meaningless output. A VAE enforces $q_\phi(z|x) \approx p(z) = \mathcal{N}(0, I)$ via the KL term, so the prior covers the latent space smoothly. Drawing $z \sim p(z)$ and decoding it yields a plausible sample from the data distribution.

A2. The ELBO: Two Forms

Algebraic equivalence. Starting from Form 1:

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_{q_\phi} [\log p_\theta(x, z) - \log q_\phi(z|x)] \\ &= \mathbb{E}_{q_\phi} [\log p(z) + \log p_\theta(x|z) - \log q_\phi(z|x)] \\ &= \mathbb{E}_{q_\phi} [\log p_\theta(x|z)] + \mathbb{E}_{q_\phi} \left[\log \frac{p(z)}{q_\phi(z|x)} \right] \\ &= \mathbb{E}_{q_\phi} [\log p_\theta(x|z)] - D_{\text{KL}}(q_\phi(z|x) \| p(z)),\end{aligned}$$

which is Form 2. $\square$

Interpretation for sensor signals. The term $\mathbb{E}_{q_\phi} [\log p_\theta(x|z)]$ is the *expected reconstruction log-likelihood*: it measures how well the decoder can recover the original sensor window from the latent code. Maximizing it pushes the model to learn a latent code that faithfully captures the signal. The term $-D_{\text{KL}}(q_\phi(z|x) \| p(z))$ is the *regularization*: it penalizes the approximate posterior for deviating from the standard Gaussian prior, preventing the model from assigning

each window an arbitrarily specific code and ensuring the latent space is well-structured and generalizes to unseen data.

A3. Why the Marginal Likelihood is Intractable

(a) When $p_\theta(x|z)$ is a neural network, $\int p(z) p_\theta(x|z) dz$ requires integrating over the entire continuous latent space $\mathbb{R}^d$. No closed form exists because the neural network introduces highly non-linear transformations of z . Numerically, quadrature becomes exponentially expensive in the latent dimension d .

(b) **Discrete z , continuous x :** $p_\theta(x) = \sum_{c=1}^K \pi_c \mathcal{N}(x; \mu_\theta(c), \sigma_\theta(c)^2)$. Exact computation requires one neural network forward pass for every value $z = c \in \{1, \dots, K\}$. For large K (e.g. $K = 2^{32}$ for a 32-bit discrete code), this is computationally infeasible.

Continuous z , discrete x : $p_\theta(x) = \int \mathcal{N}(z; 0, I) \text{Cat}(x; \pi_\theta(z)) dz$. Exact computation requires evaluating the integrand at every real-valued z with nonzero prior probability — i.e., infinitely many neural network forward passes.

A4. The Reparameterization Trick

(a) The gradient $\nabla_\phi \mathbb{E}_{q_\phi(z|x)}[f(z)] = \int (\nabla_\phi q_\phi(z|x)) f(z) dz$. The signed measure $\nabla_\phi q_\phi(z|x)$ is *not* a probability distribution: it can take negative values and does not integrate to 1. Therefore we cannot draw samples from it, and no straightforward Monte Carlo estimator exists for this integral.

(b) For $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi(x)^2))$, reparameterize as:

$$\epsilon \sim \mathcal{N}(0, I), \quad z = g(\phi, \epsilon) = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon.$$

The randomness is moved entirely into ϵ , which does not depend on ϕ . The mapping from ϵ to z is now deterministic and differentiable in ϕ .

(c) The two gradient paths are:

$$\frac{\partial z}{\partial \mu_\phi} = 1 \quad \text{and} \quad \frac{\partial z}{\partial \sigma_\phi} = \epsilon.$$

The reparameterized estimator $\nabla_\phi f(z) = \nabla_z f(z) \nabla_\phi g(\phi, \epsilon)$ has lower variance than the score-function estimator because it uses the gradient of the objective f with respect to z directly, providing an informative signal about the direction of improvement. The score-function estimator multiplies the gradient of $\log q$ by a scalar reward $f(z)$, which can vary wildly across samples, leading to high variance.

A5. Posterior Collapse

- (a) **Symptom:** the encoder ignores the input — $q_\phi(z|x) \approx p(z) = \mathcal{N}(0, I)$ for all x . Concretely, $\mu_\phi(x) \approx 0$ and $\sigma_\phi(x) \approx 1$ regardless of the content of x . The KL term $D_{\text{KL}}(q_\phi\|p) \rightarrow 0$, but reconstruction quality is poor because the decoder receives an uninformative latent code.
- (b) **Mechanism:** at the start of training, both the encoder and decoder are randomly initialized. The decoder is too weak to make use of any latent code, so the reconstruction term does not benefit from informative encodings. Meanwhile, the KL penalty *can* be reduced trivially by pushing $q_\phi \rightarrow p$. Gradient descent finds this shortcut: it minimizes the KL to zero and the decoder learns to ignore z , reconstructing only the mean of the training data.
- (c) **β -VAE / KL annealing:** set $\beta \approx 0$ initially so the KL penalty is absent; the model trains as a plain autoencoder and the decoder learns to use informative latent codes. β is then linearly increased to 1. By the time the KL penalty is fully active, the decoder already depends on z , and the encoder has no incentive to collapse.

Free Bits: replace the KL with $\sum_j \max(\lambda, D_{\text{KL},j})$, ensuring each latent dimension contributes at least λ nats of KL. This directly *forces* the encoder to use each latent unit and prevents any single dimension from collapsing to the prior.

A6. Deterministic vs. Probabilistic Inference After Training

- (a) **Downstream classifier:** *deterministic*, $z^* = \mu_\phi(x^*)$. A fixed feature vector is required; stochastic codes would introduce randomness into the classifier’s input at test time, degrading performance and reproducibility.
- (b) **Uncertainty in reconstruction:** *probabilistic*, $z^* = \mu_\phi(x^*) + \sigma_\phi(x^*) \odot \epsilon$. Multiple samples of z^* yield multiple reconstructions, whose spread quantifies the model’s uncertainty about the latent factors underlying the noisy observation. The $\sigma_\phi(x^*)$ directly encodes how uncertain the encoder is about the latent representation.
- (c) **Data augmentation:** *probabilistic*. Sampling multiple z^* from $q_\phi(z|x^*)$ and decoding each produces diverse but plausible variants of x^* , which is exactly what data augmentation requires. Deterministic encoding always produces the same reconstruction and provides no diversity.

A7. Score-Function Estimator for Discrete Latent Variables

- (a) Applying the log-derivative trick, $\partial q(z|\phi)/\partial\phi = q(z|\phi) \partial \log q(z|\phi)/\partial\phi$, the gradient becomes:

$$\nabla_\phi \mathbb{E}_{q(z|\phi)}[\log p_\theta(x|z)] = \mathbb{E}_{q(z|\phi)}[\nabla_\phi \log q(z|\phi) \cdot \log p_\theta(x|z)].$$

The term $\log p_\theta(x|z)$ is the reward signal: it weights the direction $\nabla_\phi \log q(z|\phi)$ by how well the sampled z explains the data.

- (b) The estimator is unbiased because the log-derivative trick is an exact algebraic identity. However, variance is high because the reward $\log p_\theta(x|z)$ can vary widely across samples $z \sim q$. In high-dimensional spaces, most samples contribute near-zero or very noisy gradient signals.

Part B: Mathematical Derivations

B1. Deriving the ELBO from the Marginal Log-Likelihood

Multiply and divide by $q_\phi(z|x)$ inside the log:

$$\begin{aligned}
\log p_\theta(x) &= \log \int p_\theta(x, z) dz \\
&= \log \int q_\phi(z|x) \frac{p_\theta(x, z)}{q_\phi(z|x)} dz \\
&= \log \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z|x)} \right] \\
&\geq \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z|x)} \right] \quad (\text{Jensen's inequality, } \log \text{ is concave}) \\
&= \mathbb{E}_{q_\phi} [\log p_\theta(x|z)] - D_{\text{KL}}(q_\phi(z|x) \| p(z)) = \mathcal{L}(\theta, \phi; x).
\end{aligned}$$

The gap equals the KL divergence between the approximate and true posterior:

$$\log p_\theta(x) - \mathcal{L}(\theta, \phi; x) = D_{\text{KL}}(q_\phi(z|x) \| p_\theta(z|x)) \geq 0.$$

B2. Analytic KL for Diagonal Gaussians

- (a) For $q = \mathcal{N}(\mu, \sigma^2)$, $p = \mathcal{N}(0, 1)$:

$$\begin{aligned}
D_{\text{KL}}(q \| p) &= \int q(z) \log \frac{q(z)}{p(z)} dz = \mathbb{E}_q \left[-\frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2} + \frac{1}{2} \log \sigma^{-2} \right] \\
&= -\frac{1}{2} + \frac{\sigma^2 + \mu^2}{2} - \frac{1}{2} \log \sigma^2 = \boxed{\frac{1}{2}(\sigma^2 + \mu^2 - 1 - \log \sigma^2)}.
\end{aligned}$$

- (b) For $z \in \mathbb{R}^d$ with independent coordinates:

$$D_{\text{KL}}(\mathcal{N}(\mu, \text{diag}(\sigma^2)) \| \mathcal{N}(0, I)) = \frac{1}{2} \sum_{j=1}^d (\sigma_j^2 + \mu_j^2 - 1 - \log \sigma_j^2).$$

- (c) **Check (i):** $\mu = 0, \sigma = 1$: $D_{\text{KL}} = \frac{1}{2}(1 + 0 - 1 - 0) = 0$. ✓
Check (ii): $\mu = 1, \sigma = 1$: $D_{\text{KL}} = \frac{1}{2}(1 + 1 - 1 - 0) = \frac{1}{2}$. ✓

B3. Gaussian VAE Loss Function

(a) For $p_\theta(x|z) = \mathcal{N}(x; \hat{x}_\theta(z), \sigma^2 I)$:

$$\log p_\theta(x|z) = -\frac{1}{2\sigma^2} \|x - \hat{x}_\theta(z)\|^2 - \frac{D}{2} \log(2\pi\sigma^2).$$

Taking the expectation over q_ϕ and dropping constants:

$$\mathbb{E}_{q_\phi}[\log p_\theta(x|z)] = -\frac{1}{2\sigma^2} \mathbb{E}_{q_\phi}[\|x - \hat{x}_\theta(z)\|^2] + \text{const.}$$

(b) The practical loss (negative ELBO to minimize, β -VAE generalization):

$$\mathcal{L}(x) = \underbrace{\frac{1}{2\sigma^2} \mathbb{E}_{q_\phi}[\|x - \hat{x}_\theta(z)\|^2]}_{\text{scaled MSE reconstruction}} + \beta \underbrace{D_{\text{KL}}(q_\phi(z|x) \| p(z))}_{\text{KL regularization}}.$$

Here σ^2 explicitly weights the reconstruction vs. KL balance: small σ^2 amplifies the MSE term (emphasizes reconstruction). $\beta > 1$ places extra pressure on the posterior to match the prior, encouraging disentangled representations at the cost of reconstruction quality. In many implementations σ^2 is absorbed into the loss as a fixed constant (e.g. $\sigma^2 = 1/2$ yields plain MSE), or β is re-defined to absorb both effects into a single trade-off parameter.

B4. Reparameterization: Gradient Computation

(a) With $z^{(i)} = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon^{(i)}$ and $\epsilon^{(i)} \sim \mathcal{N}(0, I)$:

$$\nabla_\phi \mathbb{E}_{q_\phi}[f(z)] \approx \frac{1}{S} \sum_{i=1}^S \nabla_\phi f(\mu_\phi(x) + \sigma_\phi(x) \odot \epsilon^{(i)}).$$

(b) By the chain rule, $\nabla_\phi f(z) = \nabla_z f(z) \cdot \nabla_\phi z$. The two gradient paths are:

$$\frac{\partial z}{\partial \mu_\phi} = 1, \quad \frac{\partial z}{\partial \sigma_\phi} = \epsilon,$$

giving the explicit gradient:

$$\nabla_\phi f(z) = \nabla_z f(z) \cdot \nabla_\phi \mu_\phi(x) + (\epsilon \odot \nabla_z f(z)) \cdot \nabla_\phi \sigma_\phi(x).$$

Both paths backpropagate through the encoder network, enabling end-to-end training with standard automatic differentiation.

B5. VAE for Time Series: Window ELBO

(a) With $p_\theta(x|z) = \mathcal{N}(\mu_\theta(z), \sigma^2 I_T)$ and $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi(x)^2))$:

$$\mathcal{L}(x; \theta, \phi) = -\frac{1}{2\sigma^2} \mathbb{E}_{q_\phi}[\|x - \mu_\theta(z)\|^2] - \frac{1}{2} \sum_{j=1}^d (\sigma_{\phi,j}^2 + \mu_{\phi,j}^2 - 1 - \log \sigma_{\phi,j}^2) + \text{const.}$$

In practice the reconstruction term uses a single sample $z = \mu_\phi + \sigma_\phi \odot \epsilon$.

(b) The anomaly score decomposes as:

$$A(x) = \underbrace{\frac{1}{2\sigma^2} \|x - \mu_\theta(z)\|^2}_{\text{reconstruction error}} + \underbrace{\frac{1}{2} \sum_j (\sigma_{\phi,j}^2 + \mu_{\phi,j}^2 - 1 - \log \sigma_{\phi,j}^2)}_{\text{KL: distance of posterior from prior}}.$$

A *healthy window* produces a small reconstruction error (the model has seen similar patterns) and a small KL (the posterior is close to the prior, indicating the window maps to a region of the latent space covered during training). An *anomalous window* produces a large reconstruction error (the decoder cannot recover it faithfully) and/or a large KL (the encoder must push q_ϕ far from $p(z)$ to find a useful representation, signalling an out-of-distribution input).

Part C: Parametric Analysis

C1. Effect of the Latent Dimension d

- (a) **Capacity:** larger d gives more latent degrees of freedom, allowing the model to represent complex breathing patterns (e.g. phase, amplitude, frequency, arrhythmia type) as separate latent factors.
- (b) **Posterior collapse risk:** as d grows, more latent dimensions are available for the encoder to ignore (collapse to the prior). Only the dimensions the decoder actually uses will be learned; the rest collapse, wasting capacity.
- (c) **Interpretability:** a small d (e.g. 2–4) allows direct visualization of the latent space, making it easy to inspect structure. Larger d requires dimensionality reduction (PCA, t-SNE) for visualization and the latent factors may no longer correspond to intuitive physiological quantities.

The practical heuristic from the lecture: start small ($d \in \{8-32\}$); select d using held-out ELBO or downstream task performance.

C2. Effect of β in the β -VAE

- (a) $\beta \rightarrow 0$: the KL term vanishes; the model trains as a plain autoencoder. Reconstruction quality is maximized but the latent space is unstructured: no prior regularization, no smooth interpolation, and no generative sampling capability.

- (b) $\beta = 1$: the standard VAE. Reconstruction and prior regularization are balanced according to the Gaussian likelihood interpretation. The latent space is smooth and can be sampled from the prior.
- (c) $\beta \gg 1$: the KL penalty dominates. The posterior is strongly compressed toward $\mathcal{N}(0, I)$, encouraging *disentanglement*: each latent dimension encodes an independent factor of variation. However, reconstruction quality degrades because the encoder cannot use informative, data-specific codes. At extreme values, posterior collapse can occur.

C3. KL Annealing Schedule

- (a) Starting with $\beta = 0$ removes the KL penalty entirely, so the model trains as a plain AE. The decoder learns to extract useful information from the latent code because there is no incentive to collapse — the gradients from reconstruction loss drive the encoder to produce informative z . Once the decoder is non-trivially dependent on z , introducing the KL gradually keeps both terms in productive tension.
- (b) If β reaches 1 too quickly, the KL penalty activates before the decoder is strong enough to use the latent code. The dynamics are equivalent to starting with $\beta = 1$ from scratch, and posterior collapse can still occur.

Annealing vs. Free Bits: KL annealing is a *temporal* control — it modulates the KL weight during training. Free Bits is a *structural* control — it enforces a minimum information content per latent dimension at all times, preventing collapse independently of the training stage. Free Bits is more robust to the choice of annealing schedule but introduces a new hyperparameter λ (bits per dimension).

C4. Gaussian vs. Bernoulli Decoder

- (a) For a Bernoulli decoder with output probabilities $\pi_\theta(z) \in (0, 1)^D$:

$$\log p_\theta(x|z) = \sum_{i=1}^D [x_i \log \pi_{\theta,i}(z) + (1 - x_i) \log(1 - \pi_{\theta,i}(z))],$$

which is the binary cross-entropy (BCE) loss, summed over dimensions.

Part D: Application and Notebook Questions

D1. Engineering Application – Breathing Anomaly Detection

- (a) The anomaly score $A(x) = -\mathcal{L}(x; \theta, \phi)$ decomposes into:
 - **Reconstruction error** $\|x - \hat{x}\|^2$: captures whether the model can faithfully recover the window. A breathing pattern unseen during training will produce a high reconstruction error because the decoder has not learned to reproduce it. This component

detects anomalies due to novel *signal morphology*.

- **KL component** $D_{\text{KL}}(q_\phi(z|x)||p(z))$: captures whether the encoder can map the window to a code within the latent region covered during training. An anomalous window may force the posterior $q_\phi(z|x)$ far from the prior, producing a high KL. This component detects anomalies due to unusual *latent representation* even when the decoder can partially reconstruct the signal.
- (b) **Healthy training:** reconstruction loss decreases monotonically; KL divergence rises gradually from near 0 (initially the encoder is close to the prior) and stabilizes at a positive value as the encoder learns to embed informative codes. **Posterior collapse:** KL falls to 0 (or near 0) while reconstruction loss stagnates at a high value — the encoder ignores the data and the decoder learns only the training mean.
- (c) **Latent dimension selection experiment:** train models for each $d \in \{2, 4, 8, 16, 32\}$ on the training split. Evaluate held-out ELBO on a validation set of *normal* windows (primary metric: higher is better). Additionally, compute anomaly detection AUC on a test set with labeled normal/anomalous windows (secondary metric). Select d that maximizes validation ELBO without sacrificing AUC. Use a chronological split (no random shuffling) to avoid look-ahead bias.
- (d) **KL annealing implementation:** at epoch t , set $\beta_t = \min(1, t/50)$; use β_t in the loss $\mathcal{L} = \text{Recon} - \beta_t \cdot \text{KL}$. Expected observations: the annealed model achieves a lower final KL (better regularization) and comparable or better reconstruction loss because the decoder learns informative codes before the KL penalty activates. Anomaly AUC may improve if the standard model suffered partial posterior collapse.

D2. Latent Space Visualization and Interpretation

- (a) A well-trained VAE should produce a *smooth, continuous* latent space where nearby points in z -space correspond to perceptually similar windows. This is enforced by the KL term, which compresses $q_\phi(z|x)$ toward $\mathcal{N}(0, I)$ for all x , preventing the encoder from assigning disjoint, isolated clusters to different training examples. In contrast, a deterministic AE can scatter codes arbitrarily with large gaps between clusters; interpolating in such gaps produces meaningless reconstructions.
- (b) **Normal windows** should cluster near the origin of the latent space (close to the prior mean) because the KL regularization encourages $q_\phi(z|x_{\text{normal}}) \approx \mathcal{N}(0, I)$. **Anomalous windows** should appear far from the origin: the encoder must push $\mu_\phi(x_{\text{anom}})$ away from 0 or inflate $\sigma_\phi(x_{\text{anom}})$ to find a useful code, both of which increase the KL distance from the prior.
- (c) Threshold selection procedure: encode all normal training windows $\{x^{(n)}\}$ to get scores $\{A(x^{(n)})\}$. Set the threshold at the $(1 - \alpha)$ -quantile (e.g. 95th or 99th percentile) of the training scores, where α is the desired false-positive rate. Justification: under the prior $p(z) = \mathcal{N}(0, I)$, well-reconstructed normal windows produce low reconstruction error and small KL; their scores cluster at low values. Setting the threshold based on training quantiles ensures that the detector’s false-alarm rate on normal data is approximately α .

References

- [1] Kingma, D.P. & Welling, M. (2013). Auto-encoding variational Bayes. *arXiv:1312.6114*.
- [2] Rezende, D.J., Mohamed, S., & Wierstra, D. (2014). Stochastic backpropagation and approximate inference in deep generative models. *ICML*.
- [3] Higgins, I. et al. (2017). β -VAE: Learning basic visual concepts with a constrained variational framework. *ICLR*.
- [4] Lucas, J. et al. (2019). Don't blame the ELBO! A linear VAE perspective on posterior collapse. *NeurIPS*.
- [5] Williams, R.J. (1992). Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8, 229–256.
- [6] Bank, D., Koenigstein, N., & Giryas, R. (2020). Autoencoders. *arXiv:2003.05991*.

Acknowledgements / Document Preparation

The problems and solutions in this document were generated with assistance from large language models (ChatGPT by OpenAI, CoPilot by Microsoft, and Claude by Anthropic), based on the course slide and notebook material. The instructor reviewed, modified, and verified all questions, solutions, and final formatting.